

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech Examination (CE/IT)
December 2013

CE703: Algorithms & Computational Complexity

Date: 03.01.2014, Friday

Time: 10:00 am to 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- (a) Determine the minimum number of comparisons needed to compute the minimum and the maximum of n numbers. 2
- (b) Why Counting Sort is called stable sort? Give atleast two applications where the property of stability is important. 2
- (c) Find total no primitive operations and growth rate of a given algorithm for best case and worst case. 3

Do Something (int A[], int n)

```
{  
    for i = 2 to n  
    {  
        index = A[i]  
        j = i - 1;  
        while (j > 0) and (A[j] > index)  
        {  
            A[j+1] = A[j]  
            j = j - 1  
        }  
        A[j+1] = index  
    }  
}
```

Q-2

- (a) Rather than separate $T[1..n]$ into two half size arrays for the purpose of merge sort, we might choose to separate it into three arrays of equal size to sort each of these recursively and then to merge the three sorted arrays. Give a formal description of this algorithm and analyse its execution time. 6
- (b) How Dijkstra's algorithm is based on greedy approach? What are the assumptions made in it? Give an example where Dijkstra's algorithm fails? Explain the reason. 4
- (c) Solve the following recurrence by using the recursion tree method. 4
 $T(n) = T(n/3) + T(2n/3) + n$

OR

- (c) What is polynomial time reducibility? Explain with the help of a suitable example. 4

Q-3 Attempt Any Two.

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- (a) List down the parameter for performance measure of following sorting algorithms and with respect to selected parameters give comparative analysis of time and space efficiency. Selection Sort, Insertion Sort and Bubble Sort.
- (b) Assuming 3-CNF satisfiability problem to be NP-Complete, prove clique problem is also NP-Complete.

- (c) Give asymptotic upper and lower bound for $T(n)$ in each of the following recurrences. Assume that $T(n)$ is constant for $n \leq 2$. Make your bounds as tight as possible and justify your answer.
1. $T(n) = 2T(n/4) + \sqrt{n}$
 2. $T(n) = 4T(n/2) + n^2 \log n$

SECTION-II

Q-4

- (a) What is running time of an algorithm? What are the factors on which running time depends on? 2
- (b) Distinguish between divide-and-conquer and dynamic programming with suitable examples 2
- (c) Prove that the running time of an algorithm is $\Theta(g(n))$ if and only if its worst case running time is $O(g(n))$ and its best case running time is $\Omega(g(n))$. 3

Q-5

- (a) Let G be a graph whose vertices are the integers 1 through 8 and let the adjacent vertices of each vertex given by the table below: 6

Vertex	Adjacent Vertices
1	2,3,4
2	1,3,4
3	1,2,4
4	1,2,3,6
5	6,7,8
6	4,5,7
7	5,6,8
8	5,7

- i) Draw G .
- ii) Order the vertices as they are visited in BFS traversal starting from vertex 1.
- iii) Derive time complexity of BFS.
- (b) Write mathematical formulation of 0-1 knapsack problem. Use dynamic programming approach to solve the following instance of the problem 5
- Maximum capacity = 11 units. Weights $[] = [1,2,5,6,7]$ Profits $[] = [1,6,18,22,28]$
- (c) Write & Explain the general recurrence equation to describe the running time of a divide and conquer algorithm. Using this equation analyze the best case and average case time complexity of quick sort. 3

OR

- (c) Let G be an undirected graph with edge costs C and T is a MST of G with respect to C . Prove or disprove that if we add 1 to all edge costs C then T is still a MST of G ? 3

Q-6 Attempt Any Two.

- (a) For the following four matrices find the order of parenthesization for the optimal chain multiplication. $A_1 = 15 \times 5$, $A_2 = 5 \times 10$, $A_3 = 10 \times 20$, $A_4 = 20 \times 25$. 14
- (b) Determine the Longest Common Subsequence of the given two string : S_1 : ALGORITHM S_2 : LOGERITHM. Show all the steps.
- (c) What is the basic idea behind String Matching? Write Rabin-Karp Algorithm for String Matching. How many valid match and spurious hits does the Rabin-Karp matcher encounter in the text $T = 2359023141526739921$ when looking for the pattern = 31415. Take value of modulo $q = 13$.